AgenticRotation: Learning Local Control for Asynchronous Visual Agents in Semantic In-Hand Reorientation

Anonymous CVPR submission

Abstract

A visual agent can decide which face of an object a user wants to see, but a dexterous hand must maintain contact while that decision is being computed. We study semantic in-hand reorientation with asynchronous visual reasoning and continuous physical execution. AgenticRotation combines a frozen RGB pose estimator, a recurrent student conditioned on a semantic goal representation, and lightweight controllers learned for pending requests and subsequent execution. The controllers use current measurements, the active goal, and request history to modulate the student's joint targets. Privileged teacher actions supervise training without entering the deployed policy. A frozen protocol evaluates the requested semantic binding and true orientation at autonomous stopping, separately from physical reach. On one marked cube with 40 scheduled trials per system, common service-eligible success rates are 74.4% for the full Astra system, 66.7% for a fine-tuned VLM without the added controllers, and 33.3% for a rule-based VLM. Learned control raises the four-arm matched estimate from 54.1% to 73.0%, although the paired difference is not statistically resolved. A separate 40-trial replication yields 70.3% for Astra and 73.0% for the fine-tuned VLM, without a resolved difference. Phase-level analysis identifies unbound inspection as the main remaining failure source. ***[TODO: Complete goal and feedback tests, additional objects, predictor comparisons, and real-robot studies before submission.]***

1. Introduction

A command such as "turn the face marked T upward" specifies a visual relation, not a sequence of joint targets. A robot must locate the mark, associate it with a controllable orientation, and move the object without losing it. The requested face may initially be hidden by the object itself or by the fingers. Moving to obtain evidence changes both visibility and contact. This coupling makes in-hand reorientation a useful setting for studying visual agents that act through physical feedback.

Large visual models provide a flexible interface between an instruction and an action. Their inference time introduces another problem. The hand cannot assume that the object remains in place while an external model reasons. Continuing the current skill can expose useful evidence or destabilize the grasp. Slowing every action can reduce movement without supplying the corrective motion needed to recover. A suitable local controller must act on the current contact state even when the next semantic decision is unavailable.

We investigate the following question: *how should a dexterous controller execute a visual agent's current intention while a new decision is pending, and resume useful motion after it arrives?* Our system separates the semantic decision rate from the motor control rate. An asynchronous agent requests inspection, binds a visible target, and updates its command from feedback. A recurrent student maps a goal representation and measured object pose to joint targets. Two compact learned controllers adapt those targets during pending requests and ordinary execution. This design permits a local policy to react without a second expensive reasoning call.

The visual and control interfaces are explicit. We adapt a ViserDex-style RGB-to-pose policy [1] to a continuous semantic goal input while reusing compatible recurrent weights. A fine-tuned local VLM directly predicts a goal vector. The Astra implementation instead emits validated public tool arguments that are mapped to the same goal space. Thus the controller consumes a common representation, but the Astra branch does not expose hidden model embeddings. The distinction makes it possible to compare a strong external agent with a compact learned grounding model without changing the deployed motor interface.

A credible evaluation must distinguish seeing the target, moving toward it, and finishing correctly. We therefore freeze the task mapping, checkpoint identities, stopping path, and physical success rule before the reported evaluation. Each system retains 40 raw trials. Service-related exclusions are applied symmetrically within a comparison and reported alongside the raw counts. Physical reach diagnostics use a separate metric. This avoids attributing a high low-level reach rate to successful semantic reasoning or autonomous stopping.

The present evidence supports a working semantic reorientation system and a specific control-learning hypoth-

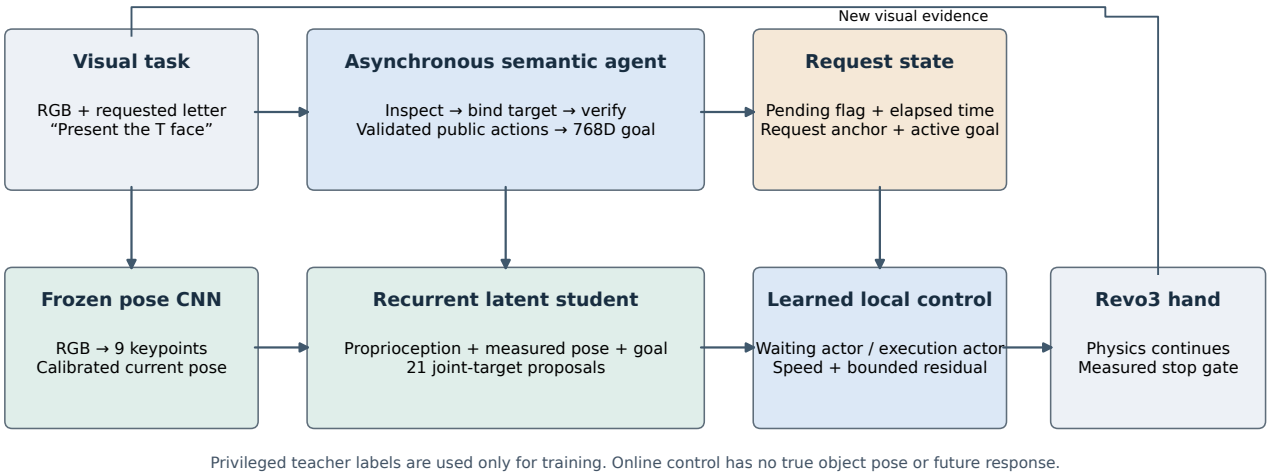


Figure 1. **Continuous control around asynchronous semantic decisions.** The visual agent selects an inspection or target skill. Its validated public command determines a semantic goal vector. A frozen RGB pose estimator and recurrent student run underneath this interface. Separate learned actors adapt joint targets while a request is pending and during execution. The hand continues moving during inference. A common measured-pose gate decides when to stop, and an independent evaluator checks true task completion. The fine-tuned VLM baseline predicts its vector directly and uses the same student without the added actors.

esis. On the frozen cube release, learned waiting and execution produce 27/37 service-eligible successes compared with 20/37 under fixed slowing and the unmodified student. The fine-tuned VLM is competitive with Astra. Current sample sizes do not establish an advantage over that baseline or a reliable interaction between the two controllers. We report these limits because they determine which experiments can substantiate the proposed method.

This draft develops three contributions: (1) a pose-mediated semantic control interface that connects asynchronous visual agents to a recurrent dexterous student, (2) phase-specific control learning from privileged teacher targets using deployable observations and active goals, and (3) a matched evaluation that separates semantic completion, contact failure, and service availability. The first cube study is complete. Multi-object transfer, causal tests of semantic conditioning, and hardware evaluation remain open experimental requirements.

2. Related Work

Visual dexterous manipulation. ViserDex uses monocular visual pose estimation and recurrent control for dexterous reorientation, with Gaussian representations supporting visual training [1]. We reuse this division between perception and control and adapt a released Revo3 policy to semantic goals. ManipTrans studies residual learning for dexterous bimanual transfer [4]. These works motivate a reusable motor policy and residual adaptation. Our residual parameterization alone is not a novelty claim. The question here is how a local policy should condition its behavior on the phase and intention of an asynchronous semantic agent.

Language and visual agents. Magma connects multimodal representations to agent actions [10]. VERA translates visual plans into robot actions and includes language-conditioned dexterous in-hand reorientation [5]. Its hand experiments include directional rotation prompts. Consequently, language-guided in-hand rotation is not an unoccupied task category. Our setting emphasizes identifying a requested marked face under occlusion, maintaining contact during external reasoning, and checking a canonical target at autonomous stopping. Those distinctions require direct evidence, rather than a claim of being the first language-controlled hand.

Asynchronous robot policies. Real-time chunk execution [2], online corrections for VLA chunks [8], and latency-aware RL [11] address inference delays through different execution and learning designs. DEFLECT studies delay-robust asynchronous VLAs through temporal counterfactual preferences [12]. LiMA combines sparse semantic intention and asynchronous action generation for dexterous manipulation [3]. These are close methodological references. Our current implementation learns small teacher-supervised controllers around a separate recurrent policy and public semantic tools. A controlled comparison with a generic time-aware residual policy is still needed to show that the active semantic goal and request anchor add value beyond reactive correction.

3. Task and Evaluation Contract

Semantic target. The current object is a cube with six marked faces. An instruction identifies one letter from $\{T, E, R, L, S, H\}$. Each face has a fixed color and a canonical world orientation $Q(g)$ that places its normal upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. This convention is disclosed and held constant for all methods. A face-normal-only score can be reported as an additional diagnostic without replacing the primary metric.

At step t, the agent observes RGB evidence I_t and the instruction ℓ. The motor system observes proprioception x_t and measured pose $\hat{p}_t$. True pose p_t, teacher actions, and future semantic responses are unavailable to online control. The simulator exposes them only for training labels and independent scoring. We denote the active goal vector by g_t, recurrent state by h_t, and the pending-request flag by m_t.

Physical time. One request may be pending at a time. Each episode allows six model calls and 3,600 logical control steps at 30 Hz. If a call starts at k and takes L wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

Autonomous completion. All systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. At stop time τ, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

Matched reporting. Every arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and instructions across semantic conditions. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

4. Method

Figure 1 summarizes AgenticRotation. The system has a semantic interface, a visual motor policy, and local adaptation. The semantic interface changes the active intention. The local controllers change how that intention is executed using current measurements.

4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a 120×120 RGB image. Calibration, unprojection, and rigid fitting recover $\hat{p}_t \in \mathbb{R}^7$, comprising position and an XYZW quaternion. The fit residual accompanies the measurement. The current rendering uses the object's Gaussian representation with live black hand occlusion. It is not a full-scene image or a real-camera observation, and no ground-truth pose fallback is used at deployment.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

The first training stage fits the new interface while preserving inherited coefficients. A second stage adapts the recurrent core using actions labeled by the frozen privileged teacher. Teacher and student rollouts are mixed in a DAgger-style procedure [7]. The frozen release uses 200 interface updates and 400 adaptation updates, each with 128 environments and 45 rollout steps, totaling 3,456,000 transitions. These stages use simulated pose noise. All reported closed-loop trials instead obtain pose from the frozen CNN. We separately measure the loss introduced by this transfer rather than assuming that initialization preserves the teacher's performance.

4.2. Semantic grounding and public actions

We maintain six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [9]. Each vector denotes

a color skill and is bound to a canonical orientation in the audited asset. The goal bank is fixed for the cube release.

Astra interface. Astra receives the requested letter and visual observations. It emits a validated public action from `inspect_face`, `reorient_target`, `verify_target`, `stop`, or `abstain`. A color argument selects a vector from B. Inspection can change the motor goal without declaring that the requested letter has been found. Inspection actions can request between 10 and 90 control steps. Subsequent visual evidence and public action history support another decision. We use the name Astra for the hosted route `gpt-6-astra` recorded in our experiment, not for a model trained here. Its exact reproducible provider version is a release requirement.

The vector delivered to the student is a local encoding of a public command. We do not access Astra's internal features, train Astra weights, or describe this pathway as end-to-end language-to-action learning. This explicit interface also enables fixed-public-plan diagnostics in which the motor system is evaluated under recorded decisions and delays.

Fine-tuned VLM. The compact baseline uses Qwen3-VL-4B-Instruct [6] with low-rank adaptation and a goal head. The head predicts a normalized continuous vector in the centered five-dimensional span of B, together with a readability probability. The continuous vector enters Eq. 3 directly. Its nearest canonical vector determines the declared binding and the orientation used by the shared stopping gate. This interface is not unrestricted open-vocabulary motor generation.

Let y indicate whether the requested mark is readable, k its skill index, and $\hat{g}$ the predicted goal. Training combines readability binary cross-entropy with positive-example goal classification and cosine alignment:

$$\mathcal{L}_{\text{ground}} = \text{BCE}(\hat{y}, y) + \mathbf{1}[y = 1]\big(\text{CE}(\hat{k}, k) + 1 - \cos(\hat{g}, g_k)\big). \tag{4}$$

The implementation averages positive terms over readable examples. Readability supervision uses geometric visibility and an image-signal filter, not human judgments. Training first uses teacher-view data and then includes observations from student trajectories. The current objective does not include an action-consistency loss.

Rule VLM. The rule baseline parses a color and readability decision from the local VLM and maps accepted colors to B. Both local baselines use the same fixed inspection sequence when a view is not accepted. They use the migrated student directly, with no learned waiting actor, execution actor, or learned completion correction. This is the primary complete-system comparison. A separate auxiliary comparison adds the same control actors to each provider to examine the control contribution.

4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector s_t from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal g_t is encoded separately. Before a binding is established it denotes the startup or inspection skill, not the unavailable future target.

A 192-dimensional sensor encoder and 64-dimensional goal encoder feed a 128-dimensional fusion layer. A waiting actor produces one speed logit r_t and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \text{sigmoid}(4r_t + \text{logit}(0.15)), \tag{5}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{6}$$

$$a_t = \text{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{7}$$

Zero outputs reproduce fixed slowing with a coefficient of 0.15 during a pending query. The environment applies its original joint mapping and smoothing. The residual gives the controller a way to correct a holding posture rather than merely suppress movement.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\text{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{8}$$

4.4. Learning execution after a decision

A second actor with the same architecture serves non-pending execution. It uses the student proposal as its base, with no speed attenuation, and learns the bounded correction in Eq. 7. Its target is the teacher action for the active goal. It can remain active throughout execution, including inspection, rather than only during a short handover interval.

Paired training examples reuse a physical state and recurrent history with different valid goals. Features and teacher labels are reconstructed for each goal. The motivation is that a generic holding action can preserve contact without moving toward the requested orientation. Goal-dependent

labels give the actor a reason to retain directional information. This data construction does not itself demonstrate that the deployed network uses semantics. A matched no-goal actor, shuffled-goal evaluation, and a same-state/different-goal intervention are required to test that mechanism.

Training also includes states collected under recorded public action and delay schedules, matching the controller's pending and execution phases more closely to deployment. The selected execution actor is the v6 candidate from development. The same weights are retained throughout the frozen evaluation. The original development selection was informed by observed development outcomes, not a blind preregistration. Reserved evaluation results did not select the checkpoint. Learned completion heads were explored during development but are not active in the reported release.

4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

This decomposition offers two distinct opportunities for learning. The low-level student can improve physical reach under its actual pose sensor. The controllers can improve how a fixed student handles a pending request or a new intention. These changes must be evaluated separately. Replacing a student changes the controller's belief and proposal distribution, so existing controller weights are checkpoint-bound and cannot be assumed to transfer unchanged.

5. Experiments

We organize evaluation around four questions: **Q1** How do complete systems compare across five objects? **Q2** What do learned control, goal conditioning, and agent feedback contribute? **Q3** Does improved visual pose prediction improve closed-loop success? **Q4** Do the findings transfer to a physical hand? Current evidence covers one cube and part of Q2. Unfinished studies are marked explicitly.

5.1. Setup and evidence scope

Experiments use the released Revo3 cube assets in an independent Isaac Lab 3.0/Newton environment. Contact capacities are validated before evaluation; earlier overflow runs are excluded. Scene details are recorded in the supplement. The original 320-trial suite contains 80 low-level diagnostics, 120 shared-control provider trials, and 120 additional Astra ablation trials. An 80-trial addendum evaluates the two bare VLM baselines. This gives 400 unique trials, including 320 semantic trials. Reusing an Astra cell across tables does not create new trials.

All semantic arms use the same scene, pose estimator, goal bank, student checkpoint, physical threshold, and shared stopping gate. The primary VLM baselines omit our learned control actors. For each arm, seeds 183101 and 183102 provide 20 trials each. Initial state matching is verified from recorded state values. Training, development, and evaluation seeds are separated, while repeated development measurements are not treated as held-out evidence. The bare-baseline addendum was declared after the earlier suite had run, following a clarification of baseline scope. It selected no new weights and changed no previous outcome.

5.2. Q1: Five-object system comparison

Table 1 reports the completed frozen cube cohort. On its common service-eligible panel, Astra succeeds in 29/39 trials, the fine-tuned VLM in 26/39, and the rule VLM in 13/39. The excluded trial was successful for both bare VLMs, which explains why their numerators also change. It would be incorrect to compare an eligible Astra rate against a raw VLM rate.

Astra and the fine-tuned VLM share 20 eligible successes. Astra has nine exclusive successes and the VLM six, giving a two-sided exact McNemar $p = 0.607$. Astra and the rule VLM share 11 successes, with 18 Astra-only and two rule-only successes, giving $p = 0.000402$. These are exploratory, unadjusted paired tests. The present evidence shows a large observed improvement over the rule baseline but does not establish an advantage over the fine-tuned model. Failure to resolve that difference is also not evidence of equivalence.

Completed independent replication. A separately registered cohort retains the same checkpoints and task criterion, with 40 trials per system. Three service-affected slots leave 37 common valid trials: Astra succeeds in 26 (70.3%), the bare fine-tuned VLM in 27 (73.0%), and the bare rule VLM in 21 (56.8%). Astra versus fine-tuned VLM has five exclusive wins and six losses ($p = 1.0$); versus rule VLM, eleven wins and six losses ($p = 0.332$). This cohort does not resolve either difference and reinforces the need to avoid a universal provider ranking. The two cohorts are reported separately, without selecting the better one or pooling them into a revised main result.

Object coverage. The present evidence concerns one marked cube. Additional object checkpoints will be integrated through the same semantic goal interface, with

System	Eligible successes	Rate (%)	95% Wilson interval	Raw successes	Raw drops
Rule VLM (bare)	13/39	33.3	[20.6, 49.0]	14/40	15
Fine-tuned VLM (bare)	26/39	66.7	[51.0, 79.4]	27/40	9
Astra + learned control (ours)	29/39	74.4	[58.9, 85.4]	29/40	10

Table 1. **Q1: frozen cube comparison.** Each system has 40 raw trials. A service exclusion in one Astra trial removes the same trial from all three methods for the eligible comparison, leaving 39. The fine-tuned and rule VLM baselines use the shared student without learned waiting or execution actors. Eligible success is the primary metric. Raw counts retain service availability for audit. Intervals describe individual eligible rates, not paired effect sizes.

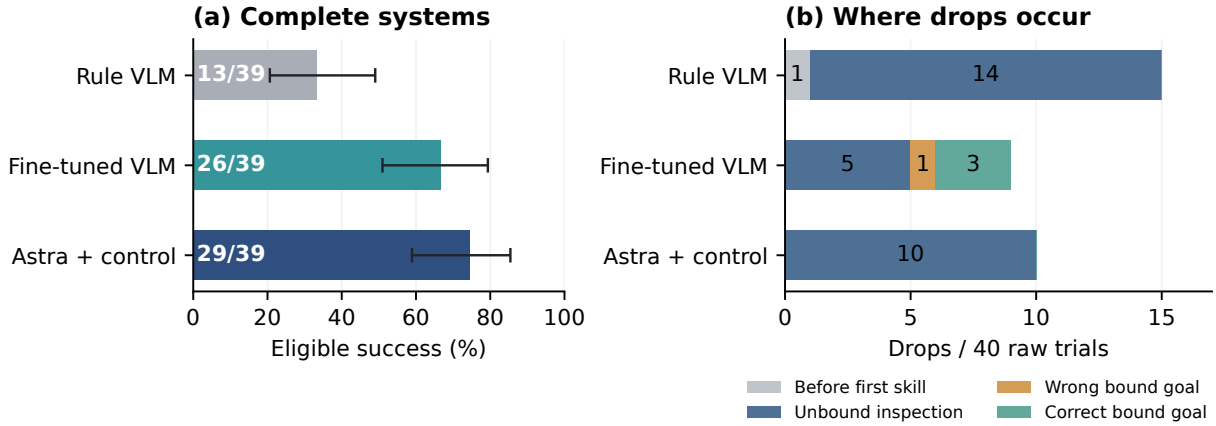


Figure 2. **Success and contact failure answer different questions.** Left: eligible success with 95% Wilson intervals on the common 39-trial panel. Right: all observed drops from 40 raw trials, classified by semantic execution context. Astra's ten drops all occur before a correct target is bound, during inspection. This concentration motivates improving evidence-seeking motion. It does not imply perfect control after binding, since reaching that phase selects a subset of trajectories.

Simulation object	Rule	Fine-tuned	Full
Cube	13/39	26/39	29/39
Object 2	TODO	TODO	TODO
Object 3	TODO	TODO	TODO
Object 4	TODO	TODO	TODO
Object 5	TODO	TODO	TODO

Table 2. **Q1: planned five-object comparison. Only the cube row is complete.** Entries use common service-eligible trials from 40 scheduled trials per arm. Raw counts and uncertainty accompany completed objects. No macro-average is computed over unfinished rows.

object-specific geometry, calibration, pose predictor, motor policy, and skill definitions recorded. An object-specific student is legitimate but does not demonstrate zero-shot object transfer. We plan approximately five objects with different grasp geometries and visual ambiguities rather than five texture variants of the same cube.

Failure structure. The ten Astra drops occur during unbound inspection. Nine occur with no model request pending and one during a pending request. None occurs before the first skill is delivered. The bare fine-tuned VLM has nine drops, two pose false stops, and two wrong-binding failures. Of its drops, five occur during unbound inspection, one after an incorrect binding, and three after a correct binding. The bare rule VLM has 15 drops, nine wrong bindings, and two horizon failures. These categories separate physical control from semantic decisions. The full method does not minimize drops among the three systems, even though it has the largest eligible success count in this cohort.

Model and control contributions. As an auxiliary comparison, we retain the original runs that use the same learned controllers for every provider. Fine-tuned VLM with shared control succeeds in 30/40 raw trials and rule VLM in 16/40, with five and twelve drops. On that panel's common 39 eligible trials, Astra and fine-tuned VLM both succeed in 29, while rule VLM succeeds in 15. Thus an unconditional ranking of Astra above the fine-tuned VLM is not supported. The main comparison assesses complete systems with the requested baseline definitions. The auxiliary panel is needed to discuss the upper-level model independently of

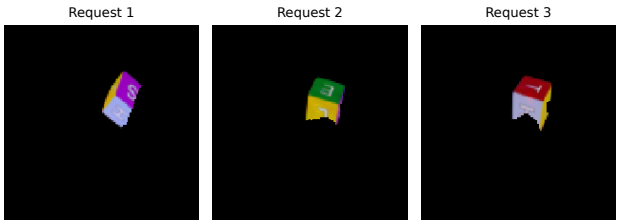


Figure 3. Actual uncropped RGB requests from the first formal Astra episode, targeting T. The episode succeeds, but the displayed request images alone do not certify final orientation. Completion is measured independently by Eq. 1.

Waiting	Execution	Success	Start	Wait	Exec.
Fixed	Student	20/37	4	0	12
Learned	Student	25/37	1	3	6
Fixed	Learned	26/37	4	0	4
Learned	Learned	27/37	0	1	9

Table 3. **Q2: matched Astra control ablation.** Every entry uses the same 37 service-eligible trials. Start, Wait and Exec. count drops before the first skill, during later pending requests, and outside waiting. The full cell reuses the original release with the factorial's common exclusions. Raw counts remain in the supplement.

exclusive controller access.

Inference and visual evidence. Astra makes 94 calls across 40 trials, with median latency 5.14 seconds and 95th percentile 7.15 seconds. One long call is service-excluded. Bare fine-tuned and rule VLM median latencies are 0.109 and 1.151 seconds. Natural latency is part of the system comparison. Usage records cover 93 calls and 154,859 tokens. Figure 3 shows actual model inputs, including hand occlusion, from one successful Astra trial; it is not a representative outcome sample.

5.3. Q2: Module and agent mechanism ablations

Table 3 holds Astra and the student fixed and varies two learned actors. Waiting control predicts speed and action residuals. Execution control predicts residuals with speed fixed to one. Fixed waiting uses coefficient 0.15, and baseline execution uses the unmodified student. Three service-affected trial identifiers are removed symmetrically, leaving 37 trials per arm. Goal conditioning and feedback are tested separately below. The measured-pose stopping gate is shared and fixed.

Learned waiting raises eligible success from 20/37 (54.1%) to 25/37 (67.6%). Learned execution with fixed waiting gives 26/37 (70.3%), and the full configuration gives 27/37 (73.0%). Full versus fixed control has paired $p = 0.210$. The point estimate improves by 18.9 percentage points, with total drops falling from 16 to 10, but reliable superiority is not established.

The components do not show a resolved interaction benefit. Adding learned waiting to execution recovery gives eight exclusive successes and seven exclusive failures, a net gain of one trial ($p = 1.0$). Startup drops fall from four to zero, while execution drops rise from four to nine. Waiting alone reduces startup drops but leaves total waiting-related drops unchanged at four. These observations motivate better coordination between inspection, waiting and execution. Different trajectories and stochastic agent decisions prevent assigning each changed failure to one actor. The completed replication has nine eligible Astra drops, one false stop and one abstention, confirming that physical failures remain a development priority.

Goal conditioning. The correct-goal, retrained no-explicit-goal and shuffled-actor-goal conditions each schedule 40 trials; their formal comparison remains incomplete. The student retains its assigned goal in every condition. A separate restored-state diagnostic verifies 19 of 20 planned states, with two assigned goals and two controller variants per state. Active-goal control gives 26/38 final in-tolerance branches and five drops, versus 29/38 and two drops without explicit actor-goal input. Both variants condition the student on the assigned goal, so action differences alone cannot identify a useful explicit actor-goal pathway. These recovery results do not establish a goal-conditioning benefit and are not semantic task success.

Feedback and visual grounding. **[TODO: Complete the same-Astra initial-plan versus feedback comparison, 40 trials per condition.]** The one-call arm emits a bounded initial plan, while the full arm may use up to six feedback calls. Both share the frozen student, actors, stopping gate and horizon. Call count and elapsed time must accompany success. Runtime preparation and development interface checks have passed, but formal feedback outcomes remain pending.

The upper-image intervention preserves predictor RGB and lighting. The bare fine-tuned VLM succeeds in 30/40 identity, 16/40 permuted and 3/40 masked trials; the bare rule VLM gives 23/40, 23/40 and 0/40. Applied-call auditing finds eleven fine-tuned trials with wrong bindings after permutation, nine matching the original symbol–color association. The rule VLM has six and three, respectively. This supports sensitivity to learned associations, without identifying an exclusive memorization mechanism. Astra's corresponding arms remain pending. Protocols and binding counts appear in the supplement.

Policy, all with frozen CNN feedback	Reach	Drops
Released pose-goal student	39/40	1
Frozen latent student	38/40	2
Latent + simulated-noise DAgger	39/40	1
Latent + CNN-feedback DAgger	40/40	0

Table 4. **Q3: completed successor development diagnostic.** Each policy uses the same 40 fresh initial states. These 160 trials are separate from the earlier 39/40 versus 35/40 frozen diagnostic and from semantic success. All physical-log validity checks pass. Predictor improvement and real-robot results remain unmeasured.

5.4. Q3: Visual pose predictor improvement

The frozen low-level diagnostic uses the same CNN for both policies. The released pose-goal student reaches before dropping in 39/40 trials, versus 35/40 for the migrated latent student, with one and five drops respectively (paired $p = 0.125$). This measures physical reach, not semantic completion.

A separate development study gives two successor students equal additional training: 400 updates and 288,000 transitions, using either simulated pose noise or frozen CNN predictions. On 40 fresh matched CNN-feedback trials, the released student reaches in 39, the frozen latent student in 38, noise-trained student in 39, and CNN-trained student in all 40 (Table 4). These small gains are not statistically resolved. This adapts the student to its sensor and does not improve the predictor. The CNN candidate remains outside the frozen semantic comparison. Training details appear in the supplement.

[TODO: Insert the colleague-provided predictor study.] Compare the original ViserDex-style predictor with the improved predictor using independent translation and rotation references, then 40 matched closed-loop trials per condition with policy and agent fixed. An optional oracle-pose diagnostic estimates control limitations and is not deployable. Any policy adaptation is a separately labeled stage. The completed student-training diagnostic above changes no predictor and is not evidence of predictor improvement. Physical superiority over a ViserDex implementation requires a matched hardware comparison.

5.5. Q4: Real-robot evaluation

Hardware comparisons should share the hand, camera, reset procedure, instructions and stopping definition. We plan 40 trials per system and object in two sessions of 20, with documented reset tolerances. An independent orientation reference must certify success to avoid evaluating the deployed predictor against its own output. Videos, model timing, joint commands and termination events should be synchronized.

[TODO: Add the physical setup photograph, object dimensions and masses, camera calibration, control frequency and latency, per-object success table, and representative failures.] The main comparisons should first stay within the original scene distribution. Lighting and background variation are useful later studies, but they do not replace evidence that the semantic controller and physical policy solve the intended task.

6. Discussion and Limitations

The results suggest that semantic inspection and contact-preserving execution must be evaluated together. In the original cohort, Astra's remaining drops occur before target binding, so holding only during inference is insufficient. The full system leads the original comparison but is one success behind the fine-tuned VLM in the independent replication. Neither cohort establishes superiority. Further comparisons must separate agent decision quality from perception and motor-control failures.

The fixed cube arrangement permits a shortcut from the requested letter to its color skill. Completed bare-VLM interventions show a fine-tuned performance loss after remapping and near-zero masked-image success. Astra intervention results remain pending. Current recovery diagnostics also do not show a benefit from explicit actor-goal input. Completing the matched goal and feedback studies is necessary to substantiate the proposed mechanism. Such interventions can retain the original background and lighting. They test the claimed mechanism rather than broad appearance generalization.

The deployed visual domain is currently a Gaussian-rendered object with hand occlusion. Simulator capacity checks establish valid execution of this chosen model, not physical fidelity or real-world robustness. A proprietary semantic service also limits exact model replication. Its public prompt, tool schema, route version, sampling settings, outputs, and measured latencies must accompany a final experimental release. No safety guarantee, broad object generalization, or statistically resolved advantage over the fine-tuned baseline is claimed from this cube study.

7. Conclusion

We presented a pose-mediated semantic reorientation system with learned control around asynchronous visual decisions. The cube study establishes a common motor interface and identifies unbound inspection as a major failure source. Learned control improves observed success over fixed control, but its interaction and target dependence remain unresolved. Matched feedback interventions, additional objects and independent hardware measurements will test whether this interface provides a repeatable advantage for agentic dexterous manipulation.

References

[1] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026. 1, 2

[2] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. *arXiv preprint arXiv:2506.07339*, 2025. 2

[3] Ning Chen, Yankai Fu, Junkai Zhao, Qianpu Sun, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. LiMA: Bridging long-term imagination to real-time dexterous manipulation via asynchronous diffusion. *arXiv preprint arXiv:2609.28431*, 2026. 2

[4] Kailin Li, Puhao Li, Tengyu Liu, Yuyang Li, and Siyuan Huang. ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6991–7003, 2025. 2

[5] Sizhe Lester Li, Evan Kim, Xingjian Bai, Tong Zhao, Tao Pang, Max Simchowitz, and Vincent Sitzmann. Turning video models into generalist robot policies. *arXiv preprint arXiv:2605.27817*, 2026. 2

[6] Qwen Team. Qwen3-VL-4B-Instruct model card. `https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct`, 2025. 4

[7] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011. 3

[8] Kohei Sendai, Maxime Alvarez, Tatsuya Matsushima, Yutaka Matsuo, and Yusuke Iwasawa. Leave no observation behind: Real-time correction for VLA action chunks. *arXiv preprint arXiv:2509.23224*, 2025. 2

[9] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 3

[10] Jianwei Yang, Reuben Tan, Qianhui Wu, Ruijie Zheng, Baolin Peng, Yongyuan Liang, Yu Gu, Mu Cai, Seonghyeon Ye, Joel Jang, Yuquan Deng, and Jianfeng Gao. Magma: A foundation model for multimodal AI agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14203–14214, 2025. 2

[11] Brian Zhu, Momen Khalil, E Harrison, Emanuele Poggi, Philipp Schmitt, Bernd Kast, Philine Meister, Pranav Atreya, Qiyang Li, Finn Ferchau, Cesar Colmenero, Yash Shahapurkar, Gokul Narayanan, Melih Erdogan, Kai Wurm, Georg von Wichert, Oier Mees, Eugen Solowjow, Andrew Wagenmaker, and Sergey Levine. Learning to act while waiting: RL finetuning of generalist robot policies under inference latency. *arXiv preprint arXiv:2608.23831*, 2026. 2

[12] Yixiang Zhu, Yonghao Chen, Jiaxiang Zou, Yusong Hu, and Xinyu Chen. DEFLECT: Temporal counterfactual preference learning for delay-robust asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. 2

LLM Usage Statement

Generative AI tools assisted with research discussion, implementation, experiment orchestration, literature retrieval, data analysis, and manuscript drafting. Astra and Qwen3-VL are also components evaluated in the research. Numerical results in this draft are derived from recorded experiments. Unfinished experiments are explicitly marked TODO. The authors must verify the final text, references, code, and results and remain responsible for the submitted work.